

Multi-Cohort Simulation Study: Shared vs. Study-Specific Factors (Proposal)

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1. Purpose and Background

Prior synthetic validation of the Supervised Bayesian Matrix Factorization (SSBMF) model was carried out only in a **single-cohort** setting. The real application pools several PDAC cohorts that differ both by **platform** (RNA-seq, microarray, proteomics) and by **biology**. In that setting the latent factors fall into two kinds:

- **Shared factors** — gene-expression programs present across *all* studies. These are the cross-cohort biological signal we want to detect and, where they are prognostic, to use for survival prediction.
- **Study-specific factors** — programs private to a *single* study. These absorb platform/batch effects and cohort-private biology that should *not* be mistaken for shared, generalizable signal.

This proposal specifies a simulation that generates data with a **known** shared / study-specific factor structure, so we can test whether the current model **natively** recovers that distinction — the same goal the previously discussed cohort dummy-indicator approach was meant to serve. If the base model already separates shared from study-specific factors, the dummy indicator columns may be unnecessary.

The design incorporates the plan from a prior meeting, with three key elements:

1. **EBMF-grounded ground truth.** Realistic gene programs are taken from an Empirical Bayes Matrix Factorization (flashier) fit to the merged real training cohorts. EBMF is also our **unsupervised benchmark** (Section 6).
2. **Shared factors carry survival signal; study-specific factors do not** ($\beta \neq 0$ on shared, $\beta = 0$ on specific). Survival is generated from the shared component only.
3. **The signal-strength of shared vs. study-specific structure is tunable** through the magnitude of the F columns, which defines the three scenarios.

2. Notation

Symbol	Meaning	Shape
C	number of cohorts (studies)	scalar
n_c	patients in cohort c ; $n = \sum_c n_c$	scalar
p	number of genes	scalar
K	total number of latent factors	scalar
$\mathcal{S}$	index set of shared factors	$ \mathcal{S} $
$\mathcal{P}_c$	index set of factors specific to cohort c	$ \mathcal{P}_c $
L	patient loadings	$n \times K$
F	gene programs (factors)	$p \times K$
β	log-hazard coefficients	K
Y	observed genomics matrix	$n \times p$

Rows of Y (and L) are **stacked by cohort**: rows $1..n_1$ are cohort 1, the next n_2 rows are cohort 2, and so on. Factors partition disjointly: $K = |\mathcal{S}| + \sum_c |\mathcal{P}_c|$.

3. Data-Generating Process

3.1 Loadings L ($n \times K$) — block structure encodes specificity

Every loading is drawn non-negative,

$$L_{ik} \sim \text{Exponential}(\text{rate} = 1), \quad \mathbb{E}[L] = 1, \quad \mathbb{E}[L^2] = 2,$$

matching the `point_exponential` prior used by the fitters. Specificity is then imposed by **zeroing out-of-cohort loadings**:

- **Shared factor** $k \in \mathcal{S}$: keep L_{ik} for *all* patients i .
- **Cohort- c -specific factor** $k \in \mathcal{P}_c$: set

$$L_{ik} = 0 \quad \text{for every } i \notin \text{cohort } c.$$

A study-specific factor therefore contributes to Y **only in its own cohort's rows** — this is the operational definition of “study-specific.” The resulting L is block-sparse: shared columns are dense, specific columns are zero outside one cohort block.

3.2 Gene Programs / Factors F ($p \times K$)

Each column $F_{\cdot k}$ is a **gene program**, normalized to unit L_2 norm so its scale is set explicitly by an amplitude (below):

- **Preferred (EBMF-grounded)**: a column of `ldf(flash_fit, type=2)$F` fit to the merged real training cohorts (realistic sparse programs), made **non-negative** (absolute value) to match the model's non-negative `point_exponential` factor prior. When K exceeds the EBMF rank, columns are recycled by index. *Matching the prior to the data sign structure is essential for a fair test*: signed programs against a non-negative prior artificially depress recovery. The EBMF benchmark is likewise run with a non-negative prior.
- **Synthetic fallback** (no real data / no flashier): for each factor pick $\lceil \text{active_rate} \cdot p \rceil$ active genes (default 5%) and set their weight to $1/\sqrt{\#\text{active}}$ (unit-norm column).

Signal-strength knob. Multiply each column by an amplitude a_k : a_{shared} for shared columns, a_{specific} for specific ones. Since the variance factor k contributes to Y is $\mathbb{E}[L_{\cdot k}^2] \|F_{\cdot k}\|^2 a_k^2$ and $\|F_{\cdot k}\|^2 = 1$, the **variance ratio scales as** $(a_{\text{shared}}/a_{\text{specific}})^2$. We call `shared_signal_ratio` = $a_{\text{shared}}/a_{\text{specific}}$ the continuous dial behind the scenarios in Section 4.

3.3 Genomics model (Gaussian)

$$Y = LF^\top + \text{offset} + E, \quad E_{ij} \sim \mathcal{N}(0, 1/\tau_j), \quad \tau_j \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 2).$$

- Per-gene precision τ_j gives gene j noise SD $1/\sqrt{\tau_j}$ ($\mathbb{E}[\tau_j] = 1$), matching the established single-cohort DGP.
- **offset** (default **0**): per the meeting notes, batch effects are absorbed by the study-specific component by construction, so an explicit offset is usually unnecessary. When enabled, one vector per cohort $o_c \sim \mathcal{N}(0, \sigma_{\text{offset}}^2 I_p)$ is added to that cohort's rows (a pure platform shift) for stress-testing.
- Y is **column-centered** before fitting (standard preprocessing).

3.4 Survival model (Weibull proportional hazards) — shared factors only

$$\eta_i = \sum_{k \in \mathcal{S}} L_{ik} \beta_k \quad (\beta_k = 0 \text{ for every specific factor}),$$

$$T_i = \left(\frac{-\log U_i}{\lambda_0 \cdot \exp(\eta_i)} \right)^{1/\alpha}, \quad U_i \sim \text{Unif}(0, 1),$$

where $\alpha = 1.5$ is the Weibull shape and $\lambda_0 = 0.01$ is the baseline scale (controls the overall event rate). The baseline cumulative hazard is $H_0(t) = (\lambda_0 t)^\alpha$. The default shared coefficients cycle the established pattern $(1.5, -1.2, 0.8, -0.5)$ to length $|\mathcal{S}|$. **Censoring** is calibrated to $\approx 30\%$ by bisection: `sensor_base` $\sim \text{Exp}(1)$, `time` = $\min(\text{event}, \text{sensor})$, `status` = $1\{\text{event} \leq \text{sensor}\}$.

Anchoring caveat (meeting notes). Because the survival outcome is linked only to shared factors, the supervision “anchors” those factors into the shared component. This is exactly the intended mechanism; the results report discusses whether it produces cleaner shared/specific separation.

4. The Three Scenarios

The scenarios differ only in how the K factors are partitioned into shared vs. specific (and, optionally, in `shared_signal_ratio`). For the primary setting $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$:

Scenario	$ \mathcal{S} $ shared ($\beta \neq 0$)	specific per cohort ($\beta = 0$)	Interpretation
All shared	6	0, 0	Studies effectively one cohort
Hybrid	2	2, 2	Both shared and private structure
Nothing shared	0	3, 3	No cross-cohort structure

Nothing-shared is a deliberate false-positive control. With $\mathcal{S} = \emptyset$, $\eta \equiv 0$ and survival carries no factor signal. The correct result is C-index ≈ 0.5 and all $|\hat{\beta}|$ below threshold — the model should *not* fabricate a shared prognostic factor. An optional knob (`specific_prognostic`, default off) can give each cohort a within-cohort prognostic factor if we later want survival signal in this regime.

Worked example — Hybrid, $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$

- **Dimensions:** $n = 300$; Y is 300×1000 , L is 300×6 , F is 1000×6 , β is length 6. Rows 1–150 = cohort 1; 151–300 = cohort 2.
- **Factor labels:** [shared, shared, specific_1, specific_1, specific_2, specific_2].
- **Loadings L :** columns 1–2 dense (all 300 patients). Columns 3–4 zero for rows 151–300 (private to cohort 1). Columns 5–6 zero for rows 1–150 (private to cohort 2). L shows a visible block-sparse pattern.
- **Coefficients:** $\beta = [1.5, -1.2, 0, 0, 0, 0]$ — only the two shared factors drive survival.
- **What a correct model should find:** factors 1–2 with loadings across *both* cohorts *and* nonzero $\hat{\beta}$; factors 3–6 cohort-private with $\hat{\beta} \approx 0$. The unsupervised EBMF benchmark should recover the same six gene programs but cannot say which two are prognostic.

Secondary sensitivity: a 3-cohort hybrid (2 shared + 1 private per cohort) and a `shared_signal_ratio` sweep within the hybrid regime.

5. Integration with the SSBMF (YFB) Model

The DGP emits exactly the inputs the fitters expect — Y ($n \times p$, column-centered), `time`, `status`, and `cohort_id` (a factor). These map directly onto:

- `fit_cox_on_yf()` — the **YFB** model, linear predictor $\eta = (YF)\beta$. This is the recommended model; `EF_norms` from the fit must be passed to `predict_cox_on_yf()` for consistent train/test scaling.
- `fit_supervised_mf_modular()` — the **LB** model, $\eta = L\beta$ (retained for comparison).

Model configuration. Both fitters use `prior_beta = "normal"` (avoids the $\beta \rightarrow 0$ collapse under YFB), `alpha = 0.5`, `lambda = 1.0`, and a `point_exponential` (non-negative) prior on L and F . Per-platform normalization — the single most important preprocessing step on real merged data — is implicitly satisfied here because every cohort block is generated on a common scale; a per-cohort-centering (“z-score”) arm is retained as a reference. The cohort dummy indicator (`cohort_id`) uses **corner-point encoding** ($C - 1$ fixed binary columns that absorb per-cohort gene means and never enter the Cox predictor).

The native-identification test. The headline analysis fits **without** the dummy indicator (`cohort_id = NULL`) and asks whether the fitted factors separate into shared vs. study-specific on their own. The $\pm$ dummy comparison then quantifies whether the indicator adds anything.

6. Supervised vs. Unsupervised (EBMF) Benchmark

EBMF (flashier, greedy + backfit) is fit to the **same simulated Y** with **no survival information**, and scored on the *identical* factor-recovery and specificity-classification metrics as the supervised fits. Because EBMF is survival-blind, it has **no β** . The comparison answers two questions:

1. Does the supervised model recover the shared/study-specific structure *as well as* the unsupervised gold standard (i.e., supervision does not hurt the factorization)?
2. Does survival supervision *additionally* yield correct β — flagging which recovered factors are prognostic — which EBMF fundamentally cannot?

This is the value-add argument for the paper: same structural recovery, plus correct survival attribution.

Note: two distinct EBMF uses. The **template-building** EBMF (Section 3.2) is fit to *real* data to manufacture realistic ground-truth gene programs. The **benchmark** EBMF here is fit to each *simulated Y* and scored as a competing method. They do not share a fit.

7. Evaluation Metrics

For each (scenario $\times$ model arm $\times$ seed) we record:

- **Shared-factor recovery** — mean best $|\text{cor}|$ between estimated and true gene programs (F columns, gene space, comparable across LB/YFB/EBMF), over the *true shared* factors.
- **Specific-factor recovery** — same, over the *true study-specific* factors.
- **Specificity-classification accuracy** — label each *estimated* factor shared/specific from its per-cohort loading energy, compare to the truth. This is the direct test of “does the model natively identify the distinction?”
- **Survival-effect recovery** — $|\hat{\beta}|$ on factors matched to true shared (expect $\$ > \$$ threshold) vs. true specific (expect ≈ 0). Reported as true-positive and **false-positive prognostic rates** (the latter is the anchoring / nothing-shared check). Threshold from `globals.yml`: `k_selection$beta_threshold`.
- **C-index** — orientation-free held-out concordance, $\max(C, 1 - C)$.

Comparisons reported: (i) YFB vs. LB; (ii) base vs. +cohort-dummy (Δ in each metric — “are dummies necessary?”); (iii) supervised vs. unsupervised EBMF.

8. Reference Data-Generating Code

The following is the complete, self-contained DGP (the project implementation wraps this same logic with configuration and seeds). A reader can run it directly given the small `.calibrate` helper.

```
# Bisection calibration of censoring scale to a target censoring fraction.
.calibrate <- function(ev, base, target = 0.30, n_iter = 40) {
  lo <- 1e-3; hi <- max(ev) * 10
  for (i in seq_len(n_iter)) {
    mid <- sqrt(lo * hi)
    if (mean(base * mid < ev) > target) lo <- mid else hi <- mid
  }
  hi
}

generate_multicohort_data <- function(
  C           = 2L,           # number of cohorts
  n_per       = rep(150L, C), # patients per cohort (rows stacked by cohort)
  p           = 1000L,        # genes
  K_shared    = 2L,           # # shared factors      (beta != 0)
```

```

K_specific = rep(2L, C),      # # specific factors per cohort (beta = 0)
F_templates = NULL,          # p x K_ebmF EBMF programs; NULL -> synthetic sparse
a_shared = 1.0,              # amplitude of shared F columns
a_specific = 1.0,            # amplitude of specific F columns
active_rate = 0.05,          # synthetic-fallback sparsity
beta_shared = NULL,          # length K_shared; NULL -> recycle c(1.5,-1.2,0.8,-0.5)
offset_sd = 0.0,             # explicit platform offset SD (0 = off)
shape = 1.5, scale0 = 0.01,  # Weibull-PH baseline
target_cens = 0.30,          # target censoring fraction
seed = 1L) {

set.seed(seed)
n <- sum(n_per)
K <- K_shared + sum(K_specific)
cohort_id <- factor(rep(seq_len(C), times = n_per))

# label + owner per column (owner 0 = shared)
labels <- c(rep("shared", K_shared),
            unlist(lapply(seq_len(C), function(cc) rep(paste0("specific_", cc), K_specific[cc]))))
owner <- c(rep(0L, K_shared),
           unlist(lapply(seq_len(C), function(cc) rep(cc, K_specific[cc]))))

# L (n x K): Exponential(1), block-zeroed outside owner cohort
L <- matrix(rexp(n * K, rate = 1), n, K)
for (k in seq_len(K)) if (owner[k] != 0L) L[cohort_id != owner[k], k] <- 0

# F (p x K): EBMF templates (unit-norm) or synthetic sparse (unit-norm)
Fm <- matrix(0, p, K)
if (!is.null(F_templates)) {
  idx <- ((seq_len(K) - 1L) %>% ncol(F_templates)) + 1L
  Fm <- F_templates[, idx, drop = FALSE]
  Fm <- sweep(Fm, 2, sqrt(colSums(Fm^2)) + 1e-10, "/")
} else {
  for (k in seq_len(K)) {
    active <- sample.int(p, max(1L, round(active_rate * p)))
    Fm[active, k] <- 1 / sqrt(length(active))
  }
}
amp <- ifelse(labels == "shared", a_shared, a_specific) # signal-strength knob
Fm <- sweep(Fm, 2, amp, "*")

# per-gene Gaussian noise: tau_j ~ Gamma(2,2)
tau <- rgamma(p, shape = 2, rate = 2)
E <- sweep(matrix(rnorm(n * p), n, p), 2, sqrt(tau), "/")

# optional explicit platform offset (off by default)
offset <- matrix(0, n, p)
if (offset_sd > 0) for (cc in seq_len(C)) {
  ov <- rnorm(p, 0, offset_sd)
  offset[cohort_id == cc, ] <- matrix(ov, sum(cohort_id == cc), p, byrow = TRUE)
}

# assemble + column-center
Y <- L %*% t(Fm) + offset + E
Y <- sweep(Y, 2, colMeans(Y), "-")

# survival from SHARED factors only
if (is.null(beta_shared))

```

```

    beta_shared <- if (K_shared > 0) rep_len(c(1.5, -1.2, 0.8, -0.5), K_shared) else numeric(0)
    beta <- numeric(K); beta[labels == "shared"] <- beta_shared
    eta <- as.vector(L %*% beta) # == 0 in the "nothing shared" scenario

    event <- (-log(runif(n)) / (scale0 * exp(eta)))^(1 / shape)
    cb <- rexp(n, rate = 1)
    cs <- .calibrate(event, cb, target = target_cens)
    ctime <- cb * cs
    time <- pmin(event, ctime)
    status <- as.integer(event <= ctime)

    list(Y = Y, time = time, status = status, cohort_id = cohort_id,
         L_true = L, F_true = Fm, beta_true = beta,
         factor_labels = labels, factor_owner = owner)
}

```

9. Open Questions for Discussion

1. **Specificity definition.** We encode study-specific factors via block-zero loadings (a factor is active in exactly one cohort). Is this the structure you intend, or should specific factors also have *non-overlapping gene programs* per cohort?
2. **Nothing-shared survival.** Default leaves survival null in the nothing-shared scenario (false-positive control). Do you prefer instead a within-cohort prognostic specific factor so each cohort has its own survival signal?
3. **Cohort count.** Primary is $C = 2$ (matches the real training data and the fixed-effects rationale); a $C = 3$ hybrid is included as sensitivity. Is 2 sufficient for the methods claim, or should 3 be primary?
4. **Signal ratio.** Default $a_{\text{shared}} = a_{\text{specific}}$. Should the hybrid scenario stress a regime where study-specific variance *dominates* shared variance (the harder, more realistic case)?

The working implementation and preliminary results are in `results/multi_cohort_sim/` (runner: `run_multicohort_sim.R`; outputs: `outputs/multicohort_sim_results.csv`). A summary results report is available at `docs/reports/multicohort_sim_resul`. Both the implementation and the results report can be updated based on feedback to this proposal.